Homework 4 Objects

Andy Mathew Tiban Camalle

Nohemi Ninibeth Tocagón Perugachi

Danilo Ismael Simba Sangucho

Tomas Patricio Tapia Arias

University of the Armed Forces – ESPE

NRC: 36863. Object Oriented Programming

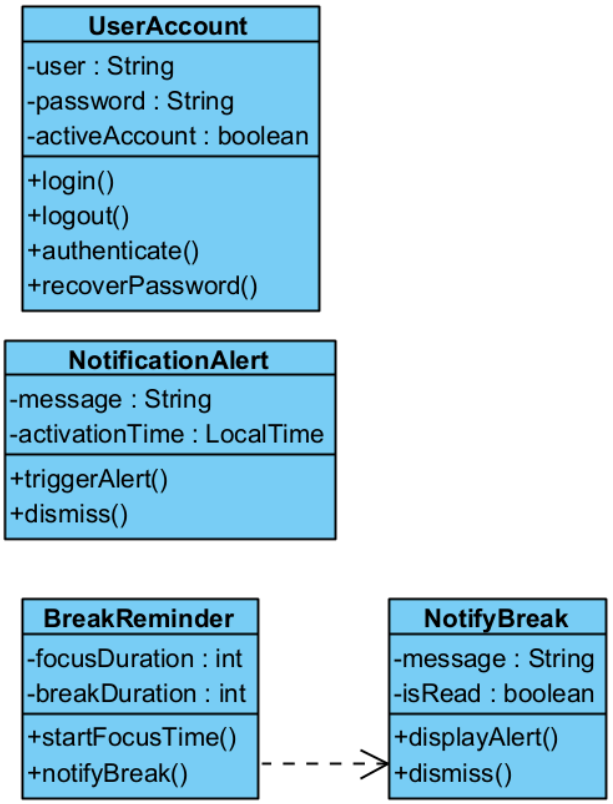
PhD Lascano Jorge Edison

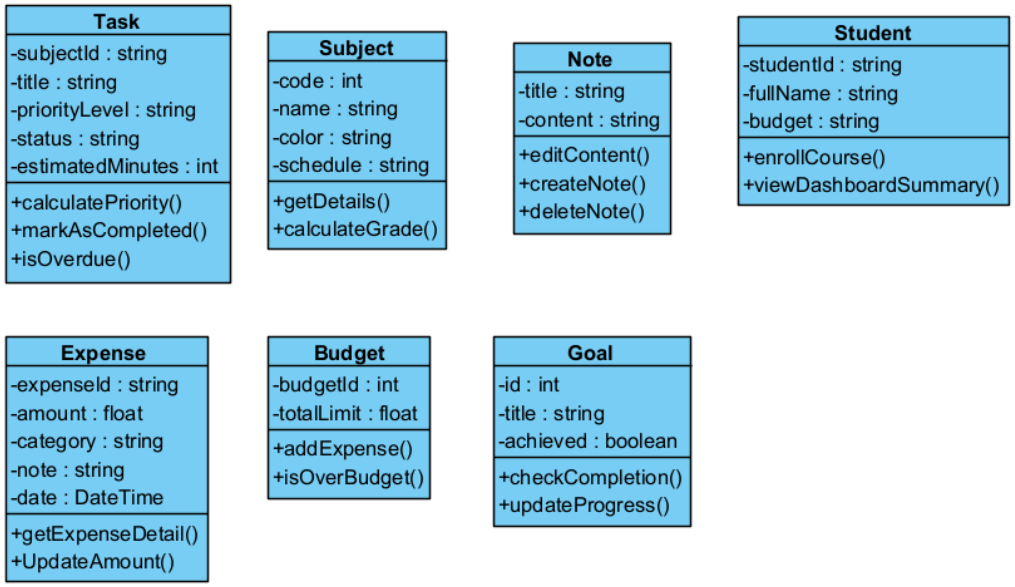
October 13, 2026

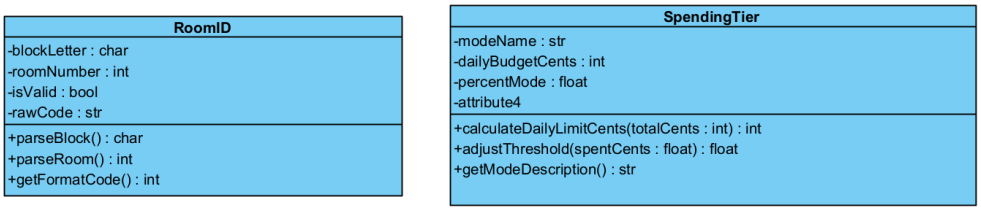
Quito – Ecuador

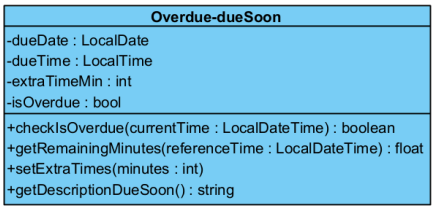
Nouns:

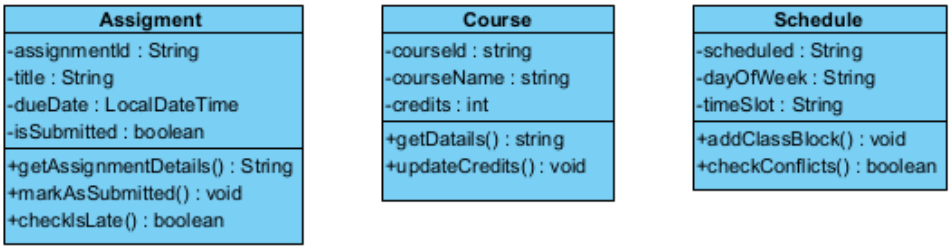
Course, Assignment, Expense, Budget, StudySession, Student, Task, Subject, Goal, Category, Tag, Schedule, Manager, Mode, Priority, Break, Reminder, Filter, User, Login, Authentication, Status, Tracking, Alert




Task
-subjectId : string
-title : string
-priorityLevel : string
-status : string
-estimatedMinutes : int
+calculatePriority()
+markAsCompleted()
+isOverdue()
Subject
-code : int
-name : string
-color : string
-schedule : string
+getDetails()
+calculateGrade()
Note
-title : string
-content : string
+editContent()
+createNote()
+deleteNote()
Student
-studentId : string
-fullName : string
-budget : string
+enrollCourse()
+viewDashboardSummary()
Expense
-expenseId : string
-amount : float
-category : string
-note : string
-date : DateTime
+getExpenseDetail()
+UpdateAmount()
Budget
-budgetId : int
-totalLimit : float
+addExpense()
+isOverBudget()
Goal
-id : int
-title : string
-achieved : boolean
+checkCompletion()
+updateProgress()


RoomID
-blockLetter : char
-roomNumber : int
-isValid : bool
-rawCode : str
+parseBlock() : char
+parseRoom() : int
+getFormatCode() : int
SpendingTier
-modeName : str
-dailyBudgetCents : int
-percentMode : float
-attribute4
+calculateDailyLimitCents(totalCents : int) : int
+adjustThreshold(spentCents : float) : float
+getModeDescription() : str


Overdue-dueSoon
-dueDate : LocalDate
-dueTime : LocalTime
-extraTimeMin : int
-isOverdue : bool
+checkIsOverdue(currentTime : LocalDateTime) : boolean
+getRemainingMinutes(referenceTime : LocalDateTime) : float
+setExtraTimes(minutes : int)
+getDescriptionDueSoon() : string


Assigment
-assignmentId : String
-title : String
-dueDate : LocalDateTime
-isSubmitted : boolean
+getAssignmentDetails() : String
+markAsSubmitted() : void
+checkIsLate() : boolean
Course
-courseId : string
-courseName : string
-credits : int
+getDatails() : string
+updateCredits() : void
Schedule
-scheduled : String
-dayOfWeek : String
-timeSlot : String
+addClassBlock() : void
+checkConflicts() : boolean